

The Blind Spot Paradox

The Blind Spot Paradox

When a self-repairing model destroys the evidence its own monitor needs

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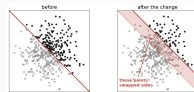
The Blind Spot Paradox: when a self-repairing model destroys the evidence its own monitor needs.

The Blind Spot Paradox

└ Concept drift: the rule changes, the data does not

First, the phenomenon. A model learns a rule: this input goes with that answer. Then the world changes and the rule no longer holds. That is concept drift. Here is the cleanest version of it, and it is what we simulate: two inputs, a straight boundary, a label on each side. The change moves the boundary, and nothing else. Only the shaded band has swapped sides, and a model trained before is now confidently wrong on it.

Concept drift: the rule changes, the data does not



A model trained on the left keeps predicting, confidently, and is now **wrong** on the shaded band.

The Blind Spot Paradox

└ Two defences, built by two separate communities

There are two defences against drift, built by two separate communities. First, a model that repairs itself: a forest of ten small decision trees, and a tree that starts failing is thrown away and replaced. Second, a watchdog: it reads the model's mistakes, adds up the excess, and calls for a human once the pile crosses a threshold. That alarm is the only thing a human ever sees.

- A **model that repairs itself**. A forest of ten small decision trees. When a tree starts failing, it is thrown away and a fresh one takes its place.
- A **watchdog that raises the alarm**. It reads the model's mistakes, adds up the excess, and calls for a human once the pile crosses a threshold.

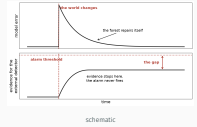
The alarm is the only thing a human ever sees.

The Blind Spot Paradox

└ Put them together, and they cancel out

Put the two together, and they cancel out. The boundary moves, the mistakes begin: that is the top curve. The watchdog needs exactly those mistakes. But the forest repairs itself, the mistakes stop, and the pile stops growing before it is high enough. The repair erased the evidence the watchdog was waiting for. The better the model fixes itself, the more surely the alarm stays silent.

Put them together, and they cancel out



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└ How often? We ran the experiment 2 000 times

How often? We re-ran the benchmark of the paper we are studying. Two thousand runs, and in each one we move the boundary at an instant we choose, so we know exactly when the change happened. The alarm fired three times out of two thousand. This is the normal case. Nobody had caught it because one community measures how fast a model recovers, the other how fast an alarm fires, and neither measured them together.

How often? We ran the experiment 2 000 times

we move the boundary at an instant we choose, then watch whether the alarm ever fires

it fired 3 times
out of 2 000 runs

- One community measures how fast a model recovers. The other measures how fast an alarm fires.
- Neither had measured them against each other.

The Blind Spot Paradox

└ Our contribution: the accepted clock is wrong

To study a race you need a stopwatch, and that is where we found something. The forest replaces its trees one at a time. The date the literature uses to say the model has recovered is the moment the very first tree is replaced: one out of ten. Wait until three quarters of the forest has been renewed, and you wait roughly ten times longer. That stopwatch does not time a repair. It times the start of one.

Our contribution: the accepted clock is wrong

called "repaired"
by the literature

86 steps

three quarters of
the forest renewed

1225 steps

0 200 400 600 800 1000 1200 1400
time after the change (steps)

median over 100 runs at $\Delta t = 0.24$, $N = 10$

It marks the first tree out of ten, not a repaired forest.

The Blind Spot Paradox

└─Where we stand

Where we stand. The paradox is real and reproducible, and we can say when the evidence falls short. The clock the field relies on is biased, always the same way, so every conclusion drawn from it inherits that bias. What remains is to watch both on one common clock, and turn this into advice a practitioner can use. Thank you.

- The paradox is **real and reproducible**, and we can say when the evidence falls short of what the alarm needs.
- The clock the field relies on is **biased, always the same way**. Every conclusion drawn from it inherits that bias.
- **Next**: watch the model and the watchdog on one common clock, and turn this into advice a practitioner can use.

Thank you.